

POINTE DE PENMARCH, France

WMO: 072000

Lat: 47.7976N

Lon: 4.3747W

Elev: 62

StdP: 14.66

Time Zone: 1.00 (EUC)

Period: 94-14

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	32.2	34.7	23.5	17.8	35.1	26.8	20.8	37.2	48.8	49.2	45.1	49.5	9.3	60	0.707

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.2	74.8	65.6	72.0	64.7	69.8	63.9	67.8	72.6	66.3	70.2	65.0	68.4	9.4	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.2	96.7	69.8	64.7	91.8	68.2	63.6	88.1	67.1	32.2	73.0	31.0	70.4	30.0	68.6	76.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 40.9	35.9	32.0	DB	28.6	81.6	4.3	4.4	25.5	84.7	23.0	87.3	20.6	89.7	17.5	92.9	(4)
(5)			WB	26.2	69.9	2.7	2.8	24.2	71.9	22.7	73.6	21.2	75.1	19.2	77.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	55.1	47.1	47.0	49.1	51.8	56.4	60.9	63.3	63.8	61.7	58.2	52.9	48.3		
	DBStd	7.44	5.62	4.45	3.82	3.79	3.88	3.62	3.56	2.89	3.32	3.72	4.23	5.80		
	HDD50	418	113	96	60	23	1	0	0	0	0	2	22	101		
	HDD65	3685	554	503	494	396	268	134	77	57	108	212	364	518		
	CDD50	2275	24	13	31	77	200	326	413	429	351	256	108	47		
	CDD65	67	0	0	0	0	2	10	25	21	9	0	0	0		
	CDH74	147	0	0	0	0	5	24	68	32	18	0	0	0		
	CDH80	21	0	0	0	0	0	3	11	4	3	0	0	0		
(14) Wind	WSAvg	15.3	17.7	17.0	15.0	15.2	14.4	13.3	13.0	12.7	12.7	15.7	18.2	18.9		
(15) Precipitation	PrecAvg	33.9	4.0	3.3	2.3	2.6	2.3	1.7	1.9	2.0	2.2	3.5	4.1	3.9		
	PrecMax	43.5	8.7	7.7	7.3	6.5	5.8	3.8	5.3	4.0	5.9	5.3	8.4	7.3		
	PrecMin	24.7	0.9	0.7	0.3	0.1	0.2	0.2	0.4	0.2	0.1	0.8	0.7	0.8		
	PrecStd	4.5	2.0	1.8	1.5	1.6	1.2	1.0	1.1	1.1	1.5	1.2	1.9	1.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.8	55.5	62.5	65.2	73.5	77.2	81.3	78.3	75.9	67.6	62.2	58.3		
		MCWB	53.9	52.2	53.1	56.7	61.6	66.1	67.6	69.3	65.8	62.7	59.8	55.8		
	2%	DB	54.8	54.0	57.3	62.2	68.9	72.3	74.8	73.2	71.6	65.2	60.3	57.1		
		MCWB	53.0	51.8	51.7	55.3	60.1	63.8	66.1	66.8	63.7	61.9	57.5	55.0		
	5%	DB	53.9	53.2	54.9	59.7	65.3	69.4	71.2	70.6	69.0	64.0	59.4	56.3		
		MCWB	52.1	51.2	51.0	54.1	58.8	62.4	65.2	65.6	63.3	61.2	56.9	54.1		
	10%	DB	53.1	52.4	53.6	57.2	62.4	66.8	68.7	68.4	67.0	62.9	58.6	55.4		
		MCWB	51.3	50.3	50.7	52.6	57.3	61.5	63.9	64.5	62.4	60.3	56.1	53.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.7	53.8	54.7	58.9	63.6	67.9	69.2	70.2	68.2	64.4	60.1	56.7		
		MCDB	55.1	54.4	59.3	63.3	70.1	74.3	75.8	74.9	73.2	66.1	61.1	57.4		
	2%	WB	53.7	52.8	53.3	56.7	61.1	65.4	67.3	68.1	66.0	62.9	58.6	55.7		
		MCDB	54.4	53.5	55.2	60.6	66.8	70.4	73.0	71.6	69.2	64.3	59.5	56.7		
	5%	WB	52.8	52.0	52.5	54.9	59.6	63.6	65.7	66.6	64.6	61.9	57.7	54.7		
		MCDB	53.6	52.8	54.1	57.9	64.3	67.9	69.9	69.7	67.5	63.2	59.1	55.9		
	10%	WB	51.8	51.0	51.7	53.7	57.9	62.0	64.4	65.1	63.4	60.8	56.5	53.7		
		MCDB	52.8	52.1	53.1	56.0	61.5	65.8	68.0	67.9	66.1	62.2	58.3	55.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.8	6.7	7.4	8.3	7.9	7.9	7.2	7.2	8.4	6.4	6.2	6.4		
		MCDBR	4.0	5.5	9.5	12.8	13.8	14.1	12.9	11.4	12.7	6.8	5.9	4.4		
	5% WB	MCWBR	4.7	5.3	7.1	8.5	8.8	9.0	7.8	7.2	8.0	5.4	5.7	5.6		
		MCWBR	3.7	4.2	6.8	9.3	11.7	11.8	10.5	9.4	9.5	5.4	4.7	4.4		
(40) Clear-Sky Solar Irradiance	taub	taub	0.340	0.352	0.377	0.386	0.389	0.386	0.384	0.374	0.362	0.362	0.342	0.333		
		taud	2.400	2.389	2.342	2.338	2.375	2.412	2.419	2.466	2.478	2.460	2.427	2.373		
	Ebn at Noon	Ebn at Noon	230	254	266	275	278	279	277	276	268	248	227	215		
		Edh at Noon	22	28	33	37	37	36	35	32	29	25	22	20		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	325	597	987	1528	1798	1949	1866	1608	1285	752	408	274		
	RadStd	RadStd	25	55	105	125	147	201	87	122	91	56	30	29		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (20 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page